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INB321G Project Paper

Thesis Statement

This project explored gene expression in bladder urothelial carcinoma (BLCA) between male lifelong non-smokers and current reformed smokers for >15 years using data from The Cancer Genome Atlas (TCGA) to identify genes that were differentially expressed based on smoking history.

Methods

Samples were identified using the GDC portal (Heath et al., 2021). Cases were selected from the TCGA Bladder Urothelial Carcinoma, TCGA-BLCA, project and were restricted to male patients with Stage IV transitional cell carcinoma originating from the bladder with open-access gene expression data available. Cases were further filtered by tobacco smoking history: one group consisted of patients classified as lifelong non-smokers, and the other consisted of patients classified as current reformed smokers for >15 years. Sample sheets for each group were exported from the GDC portal and saved as CSV files.

Gene expression count data for the selected samples were downloaded and structured in R (R Core Team, 2026) using the TCGAbiolinks package (Colaprico et al., 2016). Group 1 was set as the reference level, meaning a positive log2 fold change indicated higher expression in reformed smokers. A principal component analysis was performed to visualize sample clustering

and to identify potential outlier samples. Loci where 90% or more of the samples in either group had zero counts were removed.

Differential expression analysis was performed using the DESeq2 package (Love et al., 2014). An adjusted p-value threshold of 0.05 was used to determine statistical significance. Results were visualized using ggplot2 (Wickham, 2016) and ggrepel (Slowikowski, 2026). The complete pipeline and function code are available in Supplementary Files 1 and 2.

Results

After filtering, 34 samples were included in the analysis: 11 lifelong non-smokers (group 1) and 23 current reformed smokers for >15 years (group 2; Supplementary File 3). Principal component analysis was performed to identify potential outlier samples. PC1 and PC2 explained 9.54% and 7.91% of the variance. No samples were identified as outliers that needed to be removed (Figure 1).

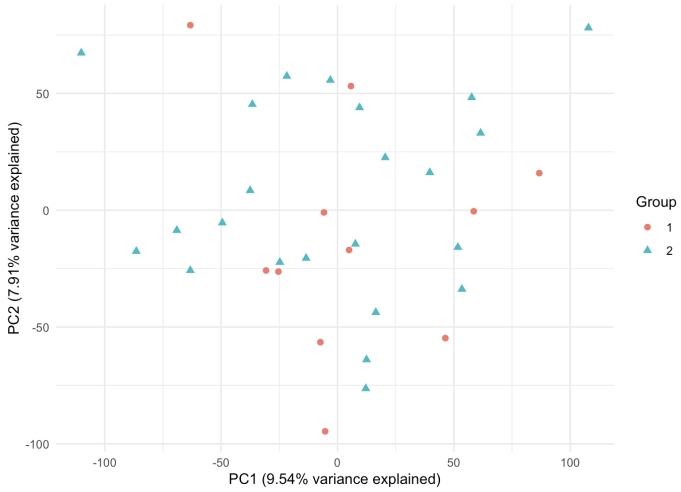


Figure 1. PCA of gene expression across 34 TCGA-BLCA samples. Pink circles represent Group 1 (lifelong non-smokers) and teal triangles represent Group 2 (current reformed smokers for >15 years). PC1 explained 9.54% of the variance, and PC2 explained 7.91% of the variance.

After locus filtering, 38,979 genes remained for analysis. 27,643 genes had non-missing p-adjusted values. A positive log2 fold change indicated higher expression in the reformed smokers (group 2) compared to the lifelong non-smokers (group 1). Using an adjusted p-value cutoff of 0.05, 82 genes were significantly downregulated, and 192 genes were significantly upregulated in reformed smokers, for a total of 274 differentially expressed genes. The top 10 most significantly differentially expressed genes from smallest to largest were SFTA2, SFTPA2, PTHLH, PEX5L, STXBP6, LINC02432, HEPHL1, FER1L6, SIGLEC5, and PRSS21. A volcano plot of all analyzed loci is shown in Figure 2. Full results for all analyzed loci are provided in Supplementary File 4.

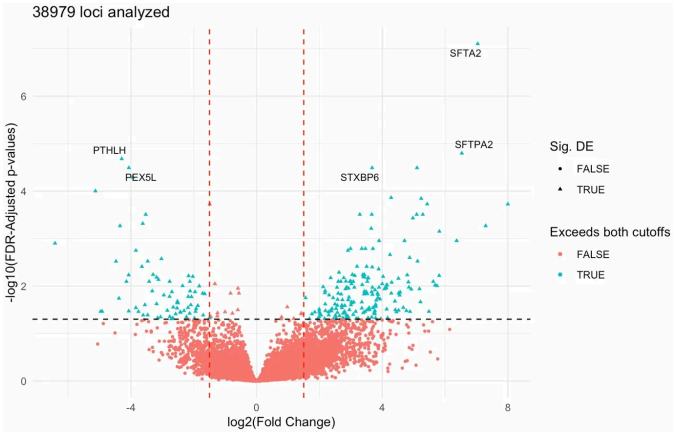


Figure 2. Volcano Plot of 38,979 analyzed loci. The x-axis shows log2 fold change, and the y-axis shows -log10 of the adjusted p-value. The triangles represent significantly differentially expressed genes (padj < 0.05), and the circles are not significant. Teal points exceed both the significance and fold change cutoffs. The horizontal dashed line marks the p = 0.05 cutoff. The vertical red dashed lines mark the log2 fold change cutoffs at -1.5 and 1.5. The top five genes meeting both cutoffs are labeled.

Supplementary Files

- Supplementary File 1: Pipeline.R
- Supplementary File 2: modifyingPipelineHw_functions.R

- Supplementary File 3: Samples.csv
- Supplementary File 4: Loci.csv

Analysis

The two most significantly differentially expressed genes in this analysis were SFTA2 and SFTPA2, which were upregulated in reformed smokers compared to non-smokers. SFTA2 encodes a hydrophilic surfactant protein that is predominantly expressed in the lung (Mittal et al., 2012). SFTPA2 encodes surfactant protein A2, one of four proteins that make up pulmonary surfactant (Floros & Tsotakos, 2021). Pulmonary surfactant is produced in the lung, where it lowers the surface tension at the air-liquid interface of the alveolus and prevents lung collapse (Floros & Tsotakos, 2021). SP-A2 also acts as part of the innate immune system, where it recognizes microbes and helps regulate inflammation (Floros & Tsotakos, 2021). SP-A levels can vary between people normally, and can also change in response to things like disease or environmental exposures (Floros & Tsotakos, 2021). This could mean that the higher expression of these surfactant genes seen in the reformed smokers' bladder tumors could mean that long-term smoking exposure changed how these genes are regulated in the bladder, rather than the bladder tissue actually becoming more "lung-like". Because smoking is the main risk factor for urothelial bladder cancer (Gabriel et al., 2011), long-term exposure could change how genes that are normally turned off outside the lung get regulated.

The third most significantly differentially expressed gene, PTHLH, was downregulated in reformed smokers compared to non-smokers. PTHLH encodes parathyroid hormone-related protein (PTHrP), which is expressed in many normal tissues, including the bladder, and has been linked to cancer progression in several types of carcinoma, including breast, prostate, pancreatic, and kidney cancer (Zhao et al., 2023). PTHrP was first identified in tumors from patients with

humoral hypercalcemia of malignancy, and has since been found to help cancer cells grow and survive (Zhao et al., 2023). Because PTHrP is associated with tumor growth, it was surprising to find it at lower levels in the reformed smoker group than in the non-smoker group. One possibility is that long-term smoking exposure reduced PTHLH expression in the bladder, and that this reduction remained in patients who had quit smoking more than 15 years before their cancer diagnosis. Beane et al. (2007) reported that some smoking-related genes in airway epithelium remained irreversibly altered in former smokers, and Gabriel et al. (2011) described both reversible and permanent smoking-derived alterations in bladder urothelium. Another possibility is that bladder cancers in non-smokers and smokers may develop through different biological pathways, so different genes might be driving the cancer in each group. Because this analysis only detects correlation, it cannot determine whether smoking history caused these differences or whether they reflect other biological differences between the groups.

Overall, the analysis was able to identify genes whose expression differed between male Stage IV BLCA patients with different smoking histories. However, there are several limitations to keep in mind. The principal component analysis showed only weak clustering by group, since PC1 and PC2 together explained less than 20% of the total variance, which suggests that most of the differences between samples were due to other factors, like tumor subtype or patient biology, rather than smoking history. The analysis only included Stage IV cases, so the results may partly come from late-stage cancer effects rather than smoking alone. Group 1 was also small (n = 11), which limited statistical power. Also, differential expression analysis only detects correlation, not causation, so these results cannot show that smoking directly caused the observed changes in gene expression. Future work could include a current-smoker group to see how gene expression changes between people who currently smoke, recently quit, and quit a long time ago, add

matched non-tumor tissue as a control, and use a larger sample size. Looking at the full list of 274 differentially expressed genes as a group, rather than just the top few, could also show which biological processes are most affected in each smoking group.

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Acknowledgments

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